

EXPERIMENT-03

AIM

To write a Python program using NumPy to compute the average sale price of houses that have more than four bedrooms from a given dataset stored in a NumPy array.

ALGORITHM

1. Start the program.
2. Import the NumPy library.
3. Load or create a NumPy array representing house data.
 - Assume columns:
0 → Bedrooms,
1 → Square footage,
2 → Sale price.
4. Apply a condition to filter houses with more than 4 bedrooms:
`house_data[:,0] > 4`
5. Use this condition to extract sale prices of only those selected houses.
6. Find the average sale price using `np.mean()`.
7. Display the result.
8. End the program.

PROGRAM

```
import numpy as np

# house_data: columns = [Bedrooms, Sq_ft, Sale_price]
house_data = np.array([
    [3, 1800, 250000],
    [5, 2400, 420000],
    [4, 2000, 310000],
    [6, 3000, 550000],
    [5, 2600, 480000]
```

```
)
```

```
# Filter for houses with more than 4 bedrooms
```

```
filtered_prices = house_data[house_data[:, 0] > 4][:, 2]
```

```
# Calculate average sale price
```

```
avg_price = np.mean(filtered_prices)
```

```
print("Average sale price of houses with more than 4 bedrooms:", avg_price)
```

SAMPLE INPUT

```
[  
  [3, 1800, 250000],  
  [5, 2400, 420000],  
  [4, 2000, 310000],  
  [6, 3000, 550000],  
  [5, 2600, 480000]  
]
```

SAMPLE OUTPUT

Average sale price of houses with more than 4 bedrooms: 483333.33

(Prices used: 420000, 550000, 480000)

RESULT

The program successfully filtered houses having more than four bedrooms and computed the average sale price using NumPy array operations.